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Project Title:	Role of high-LET radiation-induced mitotic catastrophe in mutagenesis: implication for carcinogenesis		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Comments:			
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COI Name (Institution):	Mackey, Michael (University of Iowa)		
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Task Description:	During space flight astronauts are exposed to various types of radiation and concerns have been raised regarding the genotoxic effects of such exposure. Radiation exposure in space includes components of accelerated protons and heavy ions. The principal goal of the proposed studies is to test the hypothesis that iron and proton ions of varying energies (providing high-LET values) induce mutations in exposed human cells via processes occurring in cells that undergo mitotic catastrophe (MC). MC occurs in a variety of mammalian cell lines exposed to a broad class of agents that induce a loss of regulation of cell division and it is enhanced in cells that have diminished or that lack p53 function. p53 dysfunctions are present in more than 50% of human tumors and it is well established that germline mutations and polymorphisms in the p53 gene are present in human populations and that these populations are more radiosensitive and cancer prone than the populations not carrying these traits. It is, thus, critical to acknowledge the existence of an underestimated genetically predisposed radiosensitive human population, which might potentially represent a significant		

	fraction of exposed personnel, and to estimate the risk for these personnel of developing cancer. As a first approach towards the goal of estimating cancer risk in sensitive populations, we propose here to determine the relative biological effectiveness (RBE) for the induction of MC by high-LET radiation and to determine what fraction of cells undergoing MC survives and generates clonogenic progeny. Through the completion of the three Specific Aims described in this application, we propose to test the hypotheses that MC occurring after high-LET radiation of human cells is an initiating mechanism that contributes to mutagenesis and that the yield of MC can be regarded as a predictor of carcinogenesis in vivo.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The studies conducted in this project fell in the category of risk #28 – Carcinogenesis - of the Radiation Health Bioastronautics Roadmap Table. In particular, we are addressing the following questions: 28b How can tissue specific probabilities for increased carcinogenesis risk from space radiation be best evaluated? What molecular, genetic, epigenetic, abscopal (effect that irradiation of a tissue has on remote non-irradiated tissue), and other factors contribute to the tissue specificity of carcinogenic risk? To answer this question we are testing the hypothesis that high-LET radiation leads to an enhanced incidence of MC (definition and description of MC is reported in the “Task Description” section). The idea is to define MC susceptibility of human cells exposed to densely ionizing radiation to be used as a “biomarker” for delayed effects of exposure to high-LET radiations 28i What are the probabilities that space radiation will produce DNA damage at specific sites, including clustered DNA damage? What is the likelihood that DNA damage will increase the risk of carcinogenesis? To answer this question we are testing the hypothesis that DNA damage is persistent in survivors of cells that have undergone radiation-induced-MC and that mutation induction is elevated in these survivors. 28j What mechanisms modulate radiation damage at the molecular level (e.g., repair, errors in repair, signal transduction, gene amplification, bystander effects, tissue microenvironment, etc.) that significantly impact the risk of cancers, modulate the expression of radiation damage and decrease the radiation risk, and how can the understanding of mechanisms be used to predict carcinogenic risks from space radiation? To answer this question we are following the irradiated cell populations for long time post-irradiation by means of live cell imaging. Our previous observation using the Large Scale Digital Cell Analysis System (LSDCAS) – an in-house built live cell imaging system extremely versatile – has shown that a small proportion of irradiated cells undergoing MC escape death. We are following these survivors to determine their viability. If these cells form viable colonies and are persistent for long time the risk of cancer development under these conditions can be deemed increased.
Task Progress:	This project contains three Specific Aims, in particular: Specific Aim 1: will “Determine the yield and the RBE of MC in cells exposed to iron and proton ions of varying energies”. The experiments proposed will test the hypothesis that high-LET radiation leads to an enhanced incidence of MC. Specific Aim 2: will “Determine the kinetics of cell death and the probability of cell survival in cells exposed to iron and proton ions of varying energies using live cell imaging; also determine these same parameters in the isolated cells from any surviving clones formed during live cell imaging that arise from cells that underwent radiation-induced MC”. The experiments proposed will test the hypothesis that a small proportion of irradiated cells undergoing MC escape death and form viable colonies. These surviving clones will be isolated, grown, and analyzed for mutation induction and presence of DNA damage as described in Specific Aim 3. Specific Aim 3: will “Determine the RBE for mutation induction and the presence of persistent DNA damage in cells surviving MC that were isolated during the live cell imaging experiments carried out in Specific Aim 2”. The experiments proposed will test the hypothesis that mutations and altered nucleosomal organization of chromatin (DNA damage) are persistent in survivors of MC cells thus making these cells prone to cell transformation. Experiments related to Specific Aims 1 and 2 have been carried out so far and sample processing is on going, so that only preliminary data are available at this time. Gamma-irradiation is the radiation of comparison for this project and irradiations with this source have also started and sample processing and data analysis are also on going. Under Specific Aim 1 we proposed to determine the radiation-induced cell cycle distribution/perturbation (by BrdU/PI Flowcytometry), the overproduction of cyclin B1 involved in the cell’s passage from G2 phase to mitosis (by Western Blotting and RT-PCR), and the incidence of MC (by microscopic side preparations). Experiments to analyze these end-points have been performed in May at BNL using 1 GeV iron ions irradiation and using mouse embryonic fibroblasts with functional p53 [MEF 12(1) cells] and non-functional p53 [MEF 10(1) cells, these cells are derived from the MEF12(1)]. Two gamma-irradiation experiments have also been performed (at UI) for the same end-points and with the same cells. The BrdU/PI FMC data show that changes in cell cycle phase distribution occurring after 56Fe ions irradiation of MEF cells reflect a transient delay of the cells in S and G2 phases, and that, in both cell lines, at least one cell division attempt occurs post-irradiation, with a significant fraction of cells completing this initial division. Moreover, during this time over-accumulation of cyclin B1 protein and mRNA occurs. These are all conditions that are preliminary to mitotic catastrophe induction. These results suggest that the induction of mitotic catastrophe may play a mechanistic role in the increased incidence over time of abnormal cell division in the irradiated cell populations. Under Specific Aim 2 we proposed to determine the kinetics of cell death and the probability of cell survival of irradiated cell populations using live cell imaging and the kinetics of cell death and the probability of cell survival in the isolated cells from any surviving clones formed during live cell imaging that arise from cells that underwent radiation-induced MC. We have also proposed to irradiate samples for the analysis of mutation induction at the HPRT locus and DNA damage production (via expression of gamma-HA2X) to be used as the parallel control for mutation induction and DNA damage production for the surviving clones isolated during LSDCAS imaging and that will be manipulated and analyzed according to the procedures described in Specific Aim 3. Irradiations to analyze the LSDCAS end-points have been performed in March at Loma Linda using 20 and 40 MeV proton beams at a dose of 2.0 Gy and in May at BNL using 1 GeV iron ions also at a dose of 2.0. Two gamma-irradiations have also been performed (at UI) for the same end-points and with the same cell lines. Irradiations

	for mutation induction and DNA damage production have been performed in March at Loma Linda using 20 and 40 MeV proton beams at doses of 0, 0.5, 1.0, 2.0 Gy. One gamma-irradiation has also been performed (at UI) for the same end-points and with the same cell lines. For the mutation induction experiments we have obtained the first dose response curves for the MEF 10(1) cells irradiated with gamma-rays and 56Fe ions. These preliminary data show that iron ions are about 3 times more effective than gamma-rays in inducing mutations at the HPRT locus. Mutation experiments with the proton irradiated samples are on going. For DNA damage production we have chosen to measure the expression of gamma-HA2X by Western Blotting (WB). Thus, after irradiation, cells were detached on ice and then lysed and stored at -20°. We have started the WB gamma-H2AX antibody titration using the gamma-irradiated samples. We have begun acquisition and analysis of LSDCAS image streams for the quantitation of cellular events pertaining to normal and abnormal cell division, including those events associated with MC. We have completed the design and implementation of software for manual cell event analysis; current efforts are focused on the development of automatic cell event detection software. For manual event detection, a software plugin for the graphical casViewer software has been implemented which allows for easy recording of a variety of cell events (rounding up, division, fusion, interphase death, etc.) as a function of time and field location. All identified event data are stored in an acyclic graph structure which is used for statistical analysis of the data. This manual method works well, except that event detection is quite laborious, requiring over 100 h of operator time for a moderately-sized experiment. To increase data analysis turnaround time, we have begun the development of approaches to automate the event detection process. The automatic cell event detection system starts with an accurate and rapid cell border determination, using graph-based image segmentation techniques. Subsequent processing allows the borders of individual cells to be modeled in three dimensions (two spatial and one temporal = 2D + time). The mathematical model used, based upon ring theory, can detect cell division and physiological cell death, and is being extended to provide for the characterization of specifics of the details of the cell division event (bipolar, multipolar, or failed). Analysis of the modeled image data will provide a quantitative measure of the incidence of MC-related events in treated populations; thus allowing (in a given experiment) for estimation of the percentage of cells undergoing MC and of the percentage of cells that originate from cells that have undergone MC, and that are able to form a viable colony.
Bibliography Type:	Description: (Last Updated: 02/12/2021)
Abstracts for Journals and Proceedings	Ianzini F, Kosmacek EA, Mackey MA. "A Subpopulation of Cells Undergoing Radiation-Induced Mitotic Catastrophe Escape Death by Initiating Serial Reduction Divisions: A Mechanism for Tumor Resistance, Tumor Progression and Tumor Development?" Presented at the “NASA Models of Space Radiation Risks Workshop,” Dallas, TX, March 6-7, 2007. Abstract book, “NASA Models of Space Radiation Risks Workshop,” March 2007. , Mar-2007
Abstracts for Journals and Proceedings	Kosmacek EA, Davis PJ, Ianzini F, Mackey MA. "Analysis of Cell Growth Kinetics Under Variable Conditions Using the Large Scale Digital Cell Analysis System and Cell Event Analysis Software." Presented at the “Engineering’s Research Day,” Iowa City, IA, April 19, 2007. Abstract Book, “Engineering’s Research Day,” April 2007. , Apr-2007
Abstracts for Journals and Proceedings	Ianzini F, Kosmacek EA, Symonds EA, Davis PJ, Mackey MA. "LSDCAS Studies of Cells Undergoing Reductive Division Following Radiation-Induced Mitotic Catastrophe." Presented at the “13th International Congress of Radiation Research,” San Francisco, CA, USA, July 7-12, 2007. Congress Proceedings, 13th International Congress of Radiation Research, July 2007. , Jul-2007
Abstracts for Journals and Proceedings	Ianzini F, Kosmacek EA, Mackey MA. "Radiation-Induced Cell Death Is Escaped By a Subpopulation of Cells Undergoing Mitotic catastrophe By Initiating a Program of Reduction Division: A Mechanism for Tumor Resistance, Tumor progression and Tumor Development?" Presented at the “18th Annual NASA Space Radiation Investigators’ Workshop,” Sonoma, CA, July 13-15, 2007. Abstract Book, 18th Annual NASA Space Radiation Investigators’ Workshop, July 2007. , Jul-2007
Articles in Peer-reviewed Journals	Davis PJ, Kosmacek EA, Sun Y, Ianzini F, Mackey MA. "The Large-Scale Digital Cell Analysis System: an open system for nonperturbing live cell imaging." J Microsc. 2007 Dec;228(Pt 3):296-308. PMID: 18045324 , Dec-2007
Articles in Peer-reviewed Journals	Ianzini F, Domann FE, Kosmacek EA, Phillips SL, Mackey MA. "Human glioblastoma U87MG cells transduced with a dominant negative p53 (TP53) adenovirus construct undergo radiation-induced mitotic catastrophe." Radiat Res. 2007 Aug;168(2):183-92. PMID: 17638400 , Aug-2007
Articles in Peer-reviewed Journals	Ianzini F, Bertoldo A, Kosmacek EA, Phillips EL, Mackey MA. "Lack of p53 function promotes radiation-induced mitotic catastrophe in mouse embryonic fibroblast cells." Cancer Cell International 2006 Apr;6:11. http://dx.doi.org/10.1186/1475-2867-6-11 , Apr-2006